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A Systematic Study on the Impact of the Evaporation Rate on Lithium Recovery from Brines

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Abstract

Lithium Extraction from hard brines involves a detailed and multi-stage process to maximize lithium recovery. The initial state of brine treatment is to remove or recover the divalent ion-based minerals in their pure or mixed form. Once all the divalent ions are recovered from the brine, mainly sodium and lithium ions are present in the brine with some impurities. The brine samples we have considered in this study have lithium amounts above 50 ppm. One needs to concentrate the brine using a thermal method, such as vacuum-based evaporative crystallization, to recover fresh water and crystallized salts, mainly sodium chloride. In this way, lithium concentration in the brine can be increased above 100 ppm while recovering pure NaCl salts, which have a value.

The process that is studied here involves the recovery of all the mineral salts from the brine in their purest form. We have performed a systematic study to determine the rate of evaporation at which we have the maximum amount of Lithium concentration left in the concentrated brine. In the case of solar brine from brine ponds, there has been a literature study that shows that as the brine gets concentrated due to evaporation, the crystals formed in this process are soaked with a brine that becomes richer in lithium as evaporation progresses. In the case of the hard-produced water brine, the loss of lithium to the crystals in a soaked brine is unknown. In this study, we will first perform the systematic evaporation at various evaporation cuts from 10 to 80% and map the loss of lithium to the crystallized salts.

The systematic evaporation and sample collection, and testing will allow us to arrive at the maximum evaporation rate at which the lithium loss is the lowest and lithium availability in the concentrated brine is the highest. This will allow us to maximize the lithium recovery from the evaporation process. This study will also allow us to understand the energy required to achieve the maximum lithium recovery in terms of preparing a brine ready for the direct lithium extraction process. In our case, the brine is higher than 99.5% pure Sodium Chloride with lithium and some minor impurities. The workflow presented in the study will be able to provide a template for conditioning brine for lithium recovery.

To optimize lithium extraction, it is essential to carefully manage the balance between evaporation levels and efficiency of lithium recovery, which varies from brine to brine. High evaporation levels can lead to higher lithium concentrations but may result in reduced overall recovery due to the loss of lithium in the

mother liquor. Conversely, lower evaporation levels might achieve higher recovery rates but with lower concentrations of lithium. Finding the optimal balance is key to maximizing both the concentration and recovery of lithium from the brine.

Introduction

The global transition to renewable energy and electrified transportation has elevated lithium to a critical resource, driven by its essential role in lithium-ion batteries for electric vehicles (EVs), grid-scale energy storage, and consumer electronics. In 2024, global lithium demand reached approximately 1.4 million metric tons of lithium carbonate equivalent (LCE), with projections estimating a rise to 3 million metric tons by 2030 and potentially 6 million by 2050, propelled by EV adoption and renewable energy integration [1]. This unprecedented demand has strained existing lithium supply chains, which rely heavily on conventional extraction from hard-rock spodumene and continental brine resources. Traditional brine extraction, responsible for nearly half of global lithium production, employs solar evaporation ponds that require 12–24 months and consume significant water resources (approximately 500,000 liters per ton of LCE), often disrupting local ecosystems [4]. Hard-rock mining, while faster, is energy-intensive and generates substantial waste.

These limitations have spurred the development of direct lithium extraction (DLE) technologies, which promise faster, more sustainable, and versatile lithium recovery from diverse aqueous sources.

Direct lithium extraction encompasses a suite of advanced technologies adapted from desalination and wastewater treatment, including adsorption, ion exchange, solvent extraction, membrane separation (e.g., nanofiltration and electrodialysis), and electrochemical processes. These methods selectively recover lithium from brines, geothermal fluids, or other saline solutions, achieving recovery rates of 70–90% compared to 40–60% for conventional evaporation ponds, with extraction times reduced to hours or days [3]. DLE processes leverage lithium-selective materials or processes to isolate lithium ions from complex matrices containing competing ions such as sodium, potassium, and magnesium. By enabling reinjection of processed brine into source reservoirs, DLE minimizes environmental disruption and water consumption. The versatility of DLE also allows exploitation of unconventional sources, such as geothermal brines, oilfield produced waters, and low-concentration lithium brines, thereby expanding the global lithium supply.

Commercially, DLE is gaining momentum, with production reaching 83.7 kt LCE in 2023, primarily in Argentina and China, and projected to increase to 301 kt LCE by 2028, reflecting a 2.6-fold rise in market share [1]. Over 30 companies, including Livent (Project Fenix, 100 ktpa LCE), SQM (250 ktpa LCE), CleanTech Lithium (20 ktpa LCE), and Zangge Lithium (20 ktpa LCE), are advancing DLE projects in the Lithium Triangle (Argentina, Chile, Bolivia) and China. In Europe and North America, geothermal brine projects, such as Vulcan Energy's Upper Rhine Valley initiative and E3 Metals' Clearwater project, are gaining traction. Major corporations, including ExxonMobil (Arkansas facility with a 100,000 metric ton lithium supply agreement with SK On) and Equinor (collaboration with Standard Lithium, potentially supported by a \$225 million DOE award), are investing heavily in DLE [2]. The brine (DLE) segment is forecasted to grow at a compound annual growth rate (CAGR) of 19.6% from 2025 to 2035, potentially disrupting traditional brine mining practices [3].

Despite its promise, DLE faces significant technical, economic, and environmental challenges, particularly when applied to low-concentration lithium brines (typically <200 mg/L Li). Each DLE method presents unique limitations: adsorption requires costly lithium-selective materials and high temperatures; ion exchange involves acid/base washes that increase costs and environmental impacts; solvent extraction poses risks of contamination and fire hazards; and membrane separation is less effective for high salinity or low-concentration brines [2]. For low-concentration brines, such as those found in geothermal fluids or oilfield produced waters, the challenges are amplified. The low lithium content reduces the efficiency of selective materials, as competing ions (e.g., magnesium, calcium) dominate adsorption or ion-exchange

sites, necessitating energy intensive pre-concentration steps or larger material volumes [5]. For example, brines with high magnesium-to-lithium ratios (e.g., >10:1) can reduce recovery rates by up to 30% due to competitive binding [6]. Membrane-based methods struggle with low lithium concentrations due to poor selectivity and high energy demands for driving separation, while electrochemical methods require precise voltage control to avoid coextraction of other ions, increasing operational complexity. Economically, DLE projects for low-concentration brines often have higher capital and operating costs, as larger-scale infrastructure and longer processing times are needed to achieve viable yields. Scalability remains a hurdle, with most DLE technologies still in pilot stages, requiring tailored processes for varying brine chemistries. Environmental, social, and governance (ESG) concerns also persist, as some DLE projects, particularly in Argentina's Salar del Hombres Muerto, use up to 200% more freshwater than traditional operations in Chile's Salar de Atacama [1]. Brine reinjection, while reducing surface impacts, can dilute source purity or cause subsurface contamination, raising regulatory concerns.

Looking ahead, DLE is poised to transform lithium production by addressing these challenges and unlocking new resources. Innovations in material science, such as nanostructured adsorbents and high-selectivity membranes, are expected to enhance performance for low-concentration brines by improving lithium binding efficiency and reducing interference from competing ions [5]. Hybrid systems integrating DLE with renewable energy sources, such as solar or geothermal power, could lower energy costs and carbon footprints, particularly for energy-intensive pre-concentration steps. Digitalization, including machine learning for real-time process optimization, is emerging as a tool to tailor DLE processes to diverse brine chemistries, including low-concentration sources [7]. Government incentives and clearer regulatory frameworks, especially regarding brine reinjection and water management, are likely to accelerate adoption. The potential to exploit low-concentration brines from geothermal and oilfield sources could significantly expand global lithium supply, reducing reliance on traditional high-grade deposits. As DLE technologies mature, addressing technical, economic, and ESG challenges will be critical to ensuring a sustainable and resilient lithium supply chain for the global energy transition.

Lithium loss during mechanical evaporation primarily occurs due to its adsorption or incorporation into the crystal lattices of other minerals that precipitate as the brine concentrates. Studies indicate that lithium ions (Li^+) are retained by minerals such as sodium sulfate decahydrate ($\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$), sodium chloride (NaCl), and potassium chloride (KCl). The extent of lithium loss is influenced by the adsorption capacity of these minerals, which varies based on their specific surface area and surface roughness.

- **Adsorption on Mineral Surfaces:** Research by [8] on sodium sulfate subtype Salt Lake brine found that lithium migration is restricted by adsorption on the surfaces of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$, NaCl , and KCl , with $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ exhibiting the highest adsorption capacity due to its larger specific surface area and roughness. The order of adsorption strength was $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O} > \text{NaCl} > \text{KCl}$, consistent with first-principles calculations using the VASP computational method.
- **Incorporation into Crystal Lattices:** During evaporation, lithium can be trapped within the crystal structures of precipitating salts, particularly in high-magnesium brines. A study by [9] notes that approximately 50% of the initial lithium in natural brine remains in the residual brine, attributed to retention by precipitated salts. This phenomenon complicates subsequent lithium recovery due to the high magnesium content in the residual brine.
- **Impact of Magnesium/Lithium Ratio:** The magnesium-to-lithium (Mg/Li) ratio significantly affects lithium enrichment. High Mg/Li ratios, common in many Salt Lake brines, lead to lithium co-precipitation with magnesium salts, reducing recovery efficiency. [8] observed that during sulfate-type brine evaporation, lithium enrichment is limited by sulfate mineral dispersion energy, exacerbating losses.

Several factors influence the degree of lithium loss during mechanical evaporation:

- **Brine composition:** The presence of co-existing ions such as Mg^{2+} , Na^+ , and SO_4^{2-} affects lithium behavior. For instance, high sulfate concentrations in carbonate Salt Lake brines limit lithium enrichment by promoting co-precipitation with sulfate minerals.
- **Evaporation Conditions:** The degree of brine concentration, represented by brine density, impacts lithium retention. A study on Qarhan Salt Lake brine found that lithium concentration increases with brine density until carnallite (a magnesium- potassium chloride mineral) precipitates, after which the Mg/Li ratio decreases, indicating lithium loss to solid phases.
- **Crystallization Sequence:** The sequence of mineral precipitation during evaporation determines lithium loss. For example, sodium chloride precipitates earlier, followed by potassium chloride and magnesium-containing minerals like carnallite, each potentially entrapping lithium.

Methodology

Due to chemical similarities between Mg^{++} and Li^+ , such as similar ionic radii and charge density, making selective separation is difficult. High Mg/Li ratios (often >8 , and sometimes exceeding 1000) lead to magnesium co-extraction or precipitation, reducing lithium yield and purity. Similarly, the presence of Ca^{++} competes with lithium during precipitation processes, reducing the purity of lithium carbonate (Li_2CO_3). High Ca/Li ratios require additional steps or remove calcium, often using sodium carbonate (Na_2CO_3) or oxalic acid, which increases costs and complexities [10-14].

In this study we explored the possibility of removing divalent cations such as Ca^{++} , Mg^{++} , Sr^{++} , and then brine partially to concentrate the Li^+ in the brine for lithium recovery. The objective was to first remove the divalent which typically causes co- precipitation of Lithium with the divalent ions like Mg^{++} , Ca^{++} . Once the divalent cations are removed it was hypothesized that most of the lithium will stay in the concentrated brine. Figure 1 illustrates the multi-reaction pathway, multi-stage evaporation, and direct lithium extraction process to produce minerals like $CaCO_3$, $SrSO_4$, $MgCO_3$, and Li_2CO_3 in pure forms ($>99\%$). Given the complex solubility behaviors of various chloride salts of calcium, magnesium, strontium, sodium, potassium, and lithium, traditional thermal evaporation methods are impractical for obtaining these salts in their purest forms. Our proposed method involves sequentially reacting the brine with existing cations, enabling the precipitation of individual salts in their purest forms. Once the divalent ions in the brine are fully depleted through sequential reactions, the resulting divalent-free brine requires up to 30% less energy for the multi-stage evaporation process. This process yields fresh water and a concentrated brine with Na^+ , K^+ , and Li^+ monovalent cations. With the initial Li^+ concentration of over 50 ppm in the original brine, the concentrated brine can reach Li^+ levels exceeding 150+ ppm after 70% evaporation, enhancing the viability and profitability of the DLE process.

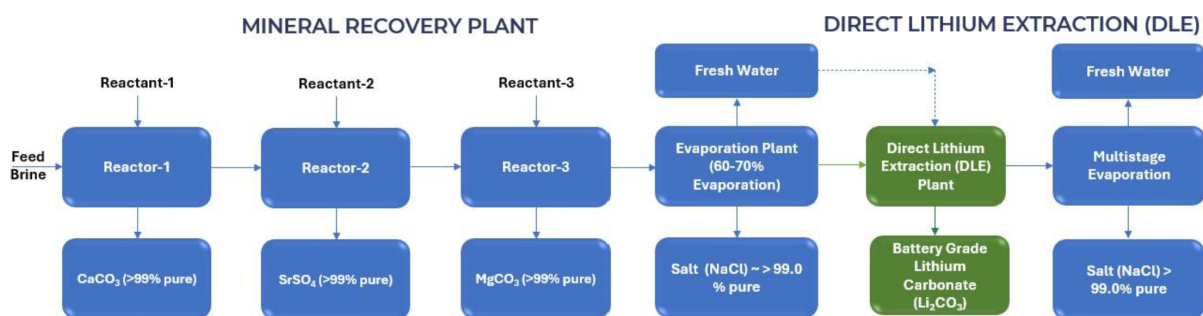


Figure 1—Zero mineral and liquid discharge (ZMLD) process flow diagram.

In the traditional direct lithium extraction process, over 50% of the capital and energy costs are dedicated to removing divalent cations from the brine before extracting the Li^+ . Our process, however, not only produces multiple salts in their pure forms but also recovers a divalent-free brine. This allows us to recover Li^+ up to 30% less capital and operating costs compared to conventional direct lithium extraction methods. After extracting Li^+ from the concentrated brine, the brine can be further dried to recover pure sodium chloride.

Laboratory experiments were performed using the field brine with Li^+ concentration above 50 ppm with very high concentrations of Ca^{++} , Mg^{++} , Sr^{++} , and Na^+ (see Figure 2). This particular brine is a high-salinity brine with significantly high hardness. A total of 10 L of brine was used to follow the reaction sequence, where most of the divalent ions are removed by precipitating each divalent salt, such as CaCO_3 , SrSO_4 , and MgCO_3 , individually in their purest form with > 99% purity. The brine, which is divalent-free, is then evaporated in different evaporator cuts, and the lithium concentration in the concentrated brine after each evaporation cut is measured. Figure 2 shows the reaction sequence and quantities of solids recovered from each step.

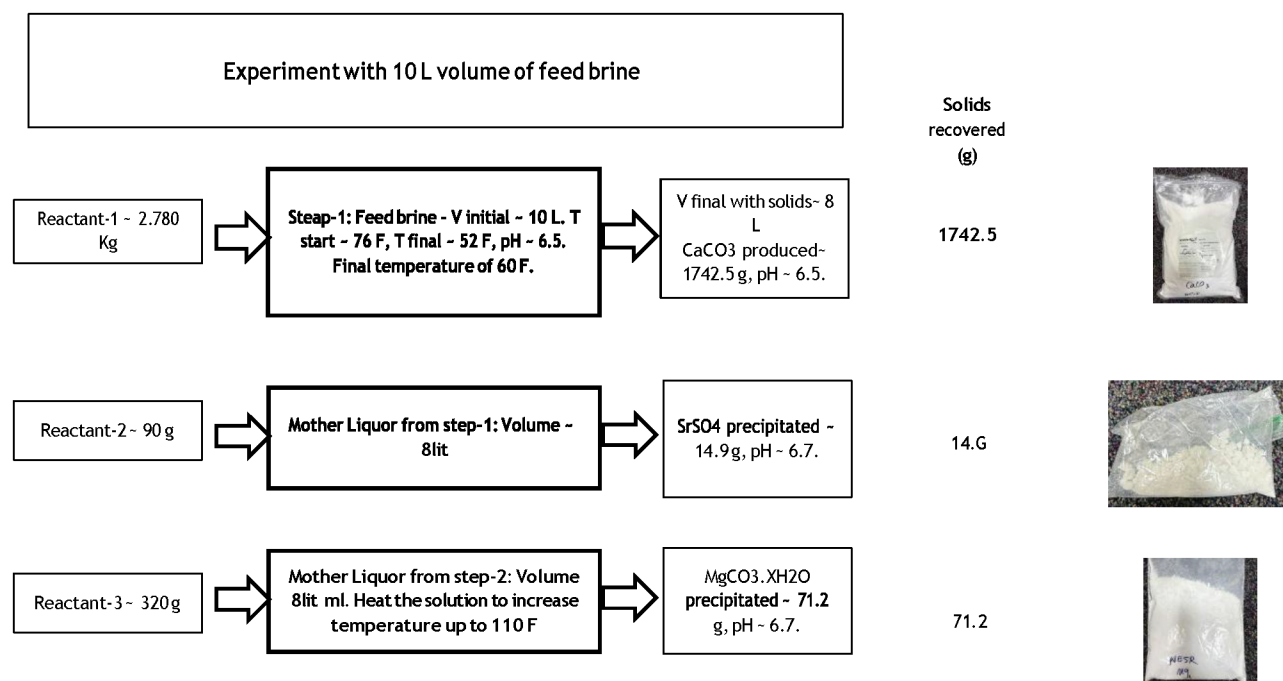


Figure 2—Laboratory experiment to precipitate the divalent cations as solid salts in sequence to produce CaCO_3 , SrSO_4 , and MgCO_3 .

Figure 3 presents the measured lithium concentrations in the concentrated brine following each evaporation stage, using Inductive Coupling Plasma Mass Spectroscopy (ICP-MS) equipment. The ICP-MS was calibrated by measuring lithium standards at concentrations of 10 ppm, 50 ppm, 100 ppm, 200 ppm, and 1000 ppm to ensure high measurement accuracy. The red curve illustrates the estimated Li^+ concentration in the concentrated brine, based on volume reduction due to evaporation, under the assumption that 100% of lithium remains in the concentrated brine after each evaporation stage. The blue curve represents the measured Li^+ concentration in the concentrated brine at each evaporation stage.

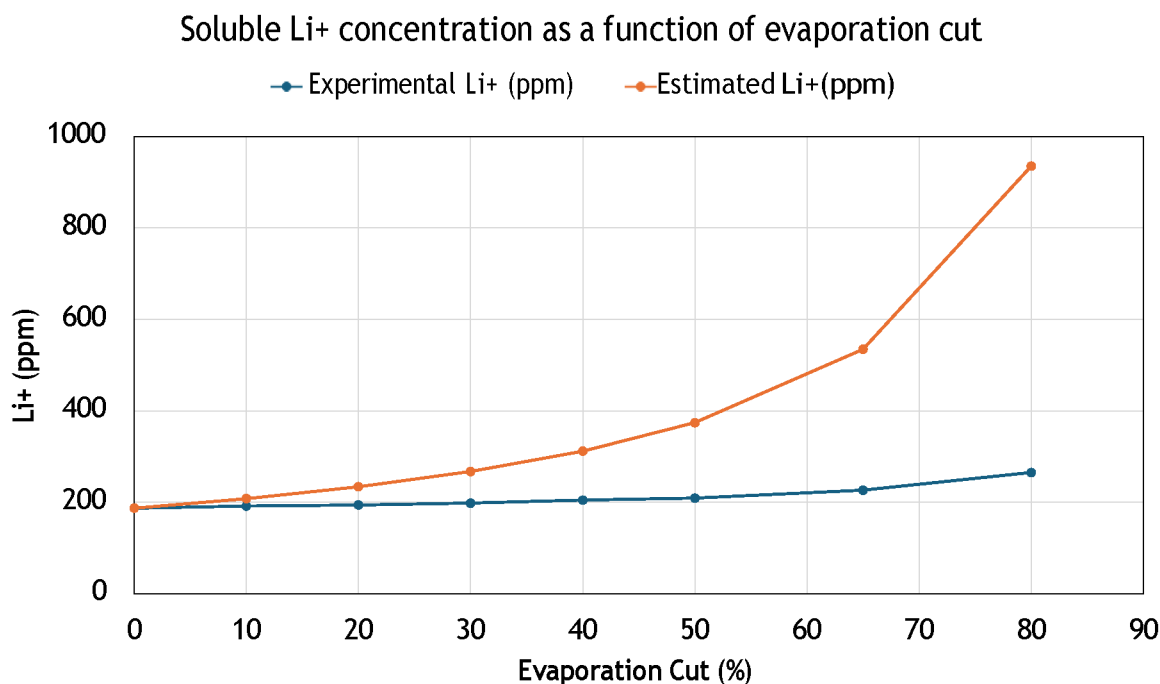


Figure 3—Concentration of lithium in the concentrated brine plotted as a function of evaporation cut%.

A significant discrepancy between the estimated and measured lithium concentrations is observed, primarily attributed to substantial lithium loss to the crystallized salt, predominantly sodium salt, with possible co-precipitation of lithium with sodium salt. It is essential to highlight a critical aspect of this process: during the initial reaction sequence for precipitating Ca⁺⁺, Mg⁺⁺, and Sr⁺⁺ salts, sodium salts are used as reactants, resulting in a filtrate or mother liquor enriched in sodium concentration after each reaction. For high salinity and hard brine, upon completion of the third reaction, the brine approaches the saturation limit of sodium chloride, with the predominant salt being >98% sodium and a minor presence of K⁺ and Li⁺. Consequently, once evaporation of the divalent-free brine begins after the third reaction, crystallization of solids can commence even at early evaporation stages like 10%. Several literature sources have discussed the co-precipitation of lithium with divalent ions such as Ca⁺⁺ and Mg⁺⁺, however, there is limited research available regarding the co-precipitation of lithium with monovalent salts such as sodium chloride.

Table 1—Purity of the salts precipitated from the sequential chemical reactions of batch 1

CaCO ₃	Elements	%	SrSO ₄	Elements	%	MgCO ₃	Elements	%
	CaCO ₃	GG.76		SrSO ₄	GG.78		MgCO ₃	GG.76
	Al	0.011		Al	0.002		Al	0.003
	Si	0.009		Cl	0.013		Si	0.001
	Cl	0.015		Ca	0.102		S	0.101
	Mg	0.057		Na	0.008		Ca	0.035
	Na	0.011		Fe	0.003		Na	0.02
	Fe	0.012		Ba	0.011		Fe	0.004
	Sr	0.115		Si	0.001		K	0.024
	K	0.012		Mg	0.038		Sr	0.055
				K	0.042			

Table 2—Purity of the salts precipitated from the sequential chemical reactions of batch 2.

CaCO ₃	Elements	%	SrSO ₄	Element	%	MgCO ₃	Elements	%
	CaCO ₃	GG.76		SrSO ₄	GG.80		MgCO ₃	GG.76
	Al	0.011		Al	0.001		Al	0.003
	Si	0.009		Cl	0.021		Si	0.001
	Cl	0.015		Ca	0.098		S	0.101
	Mg	0.057		Na	0.012		Ca	0.035
	Na	0.011		Fe	0.005		Na	0.02
	Fe	0.012		Ba	0.009		Fe	0.004
	Sr	0.115		Si	0.001		K	0.024
	K	0.012		Mg	0.022		Sr	0.055
				K	0.032			

Figures 3 and 5 clearly show the potential losses of lithium due to its interaction with sodium crystals. The extraction of lithium from brine through mechanical evaporation is a critical process for meeting the growing demand for lithium-ion batteries. However, significant losses of lithium occur due to its interaction with sodium salt crystals, such as sodium chloride (NaCl) and sodium sulfate decahydrate ($\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$), which precipitate during evaporation. Lithium loss to sodium salt crystals is driven by adsorption and incorporation into crystal lattices during the evaporation process:

- **Adsorption on Crystal Surfaces:** Studies show that lithium ions (Li^+) are adsorbed onto the surfaces of sodium salts. Wang et al. (2022) found that in sodium sulfate subtype Salt Lake brine, $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ exhibits a higher adsorption capacity for lithium compared to NaCl, attributed to its larger specific surface area and surface roughness. The adsorption strength follows the order $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O} > \text{NaCl}$, as confirmed by first-principles calculations using the VASP method.
- **Incorporation into Crystal Lattices:** As evaporation concentrates the brine, lithium can be entrapped within the growing crystal structures of NaCl and $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$. Klimko et al. (2025) noted that approximately 50% of lithium in natural brine remains in the residual phase, with a portion lost to precipitated sodium salts, particularly in high-magnesium brines where co-precipitation is prevalent.
- **Impact of Evaporation Dynamics:** Figure 4 highlights a discrepancy between experimental (stable at 200-300 ppm) and estimated (rising to over 800 ppm) lithium concentrations. This suggests that significant losses to sodium salts occur as evaporation progresses beyond 50%. It is known that the rate of crystallization of sodium salt increases as the brine becomes highly concentrated.

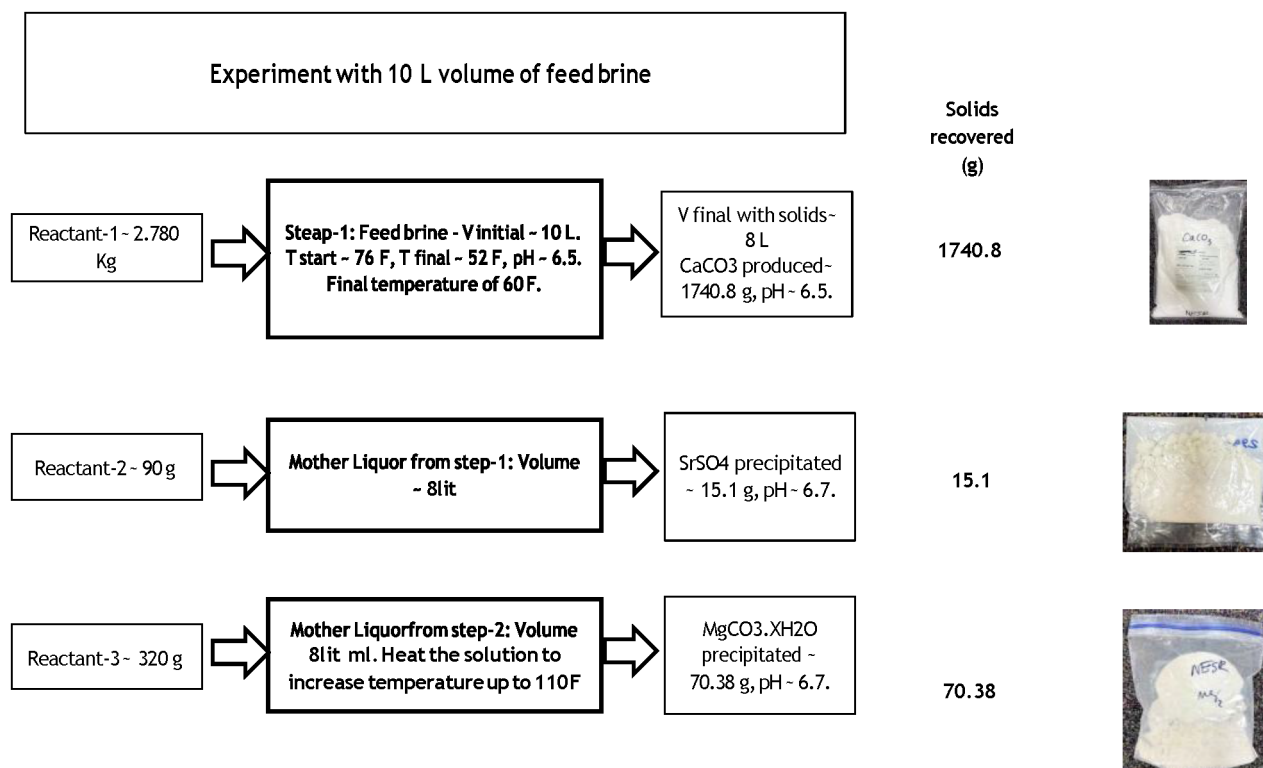


Figure 4—Laboratory experiment to precipitate the divalent cations as solid salts in sequence to produce CaCO₃, SrSO₄, and MgCO₃.

Soluble Li⁺ concentration as a function of evaporation cut

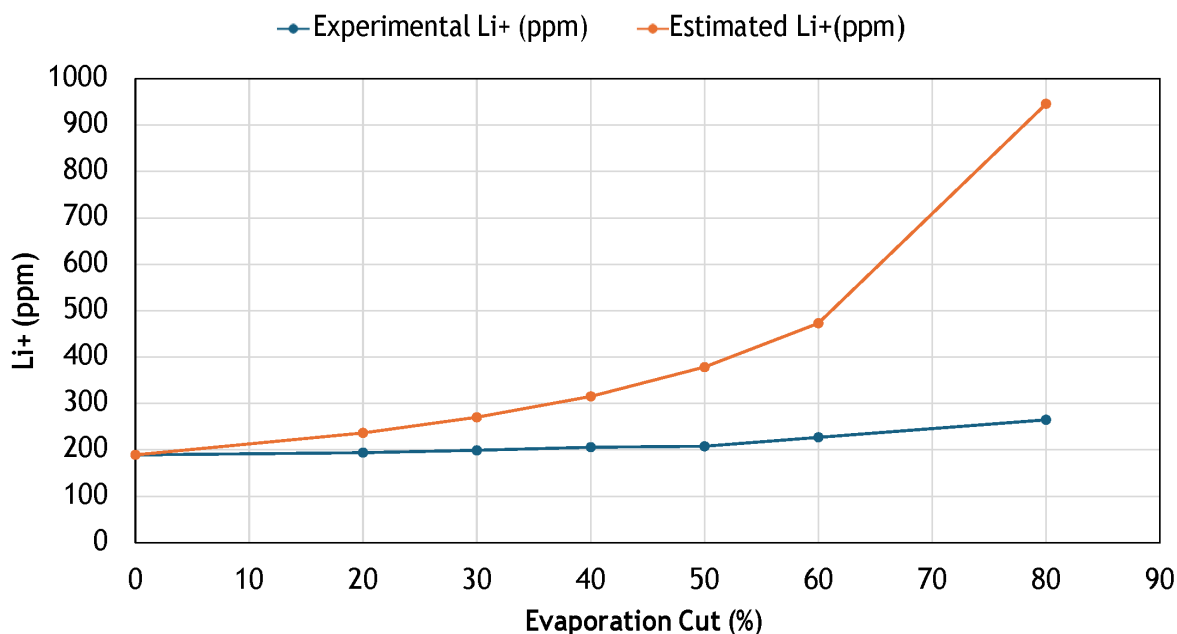


Figure 5—Concentration of lithium in the concentrated brine plotted as a function of evaporation cut%.

Several factors contribute to the extent of lithium loss to sodium salt crystals:

- **Brine Composition:** High concentrations of sodium and sulfate ions enhance lithium adsorption and co-precipitation. In sulfate- enriched brines, the presence of Na₂SO₄·10H₂O increases lithium retention due to its favorable crystal surface properties.

- Evaporation Conditions: The rate and degree of evaporation affect crystal growth. Slower evaporation may increase adsorption time, while rapid evaporation can lead to more lithium entrapment in crystal lattices.
- Mg/Li Ratio: A high magnesium-to-lithium ratio, common in many brines, exacerbates lithium loss by promoting the formation of mixed precipitates, including sodium-magnesium-lithium complexes. In the case of the brine in this study, due to preconditioning, the brine used in evaporation experiments is almost calcium and magnesium-free. It is demonstrated that even without magnesium interference, a significant amount of lithium can be lost to the sodium crystals.

Having a preconditioned brine that is divalent-free is critical for the lithium recovery process. It will certainly improve the recovery of direct lithium extraction using adsorbents and ion exchange methods. However, there are two competing factors here:

- Once the brine is divalent-free, the Mg/Li and Ca/Li ratio is almost zero. This will improve the selectivity of Li^+ extraction.
- In case of low concentration Li^+ brine, before considering evaporation for the divalent-free brine, it is critical to understand the lithium loss during evaporation, which can be managed by removing sulfate and keeping evaporation cut on the lower side.
- some membranes can separate the Na^+ and Li^+ from the brine, which is predominantly Na^+ and Li^+ . One needs to adopt membrane technology instead of using evaporation-based technology to separate the Li^+ from the brine. This will avoid significant losses of lithium during the lithium extraction process.

In this paper, we have discussed that the presence of magnesium and calcium in brines significantly impacts lithium recovery using any DLE process. Magnesium competes for adsorption sites, forms complexes with lithium, and causes fouling, while calcium reduces selectivity, leads to scaling, and complicates desorption. These effects can reduce lithium recovery by 10-30% in high-Mg/Ca brines. We have demonstrated that mitigation strategies, including pre-treatment and process optimization, are essential to improve efficiency. Addressing these challenges through further research and development will be crucial for scaling DLE technologies to meet global lithium demand sustainably. The low concentration Li^+ brines are gaining lots of interest to meet the demand for lithium, which requires an integrated process that improves selectivity & recovery efficiency of lithium. We propose a process sequence that could be beneficial in lithium recovery while recovering other minerals in the pure form, which will make the brine divalent-free for lithium extraction.

Conclusions

The global transition to renewable energy and electrified transportation has underscored the critical importance of lithium, with demand soaring to 1.4 million metric tons of lithium carbonate equivalent (LCE) in 2024 and projected to reach 3 million by 2030 and 6 million by 2050 due to the rise in electric vehicle (EV) adoption and energy storage needs. Traditional lithium extraction methods, such as mechanical evaporation from brines and hard-rock mining, are proving inadequate, with evaporation methods losing up to 50% of lithium to sodium salt crystals like NaCl and $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ due to adsorption and incorporation, as evidenced by the discrepancy between experimental (200-300 ppm) and estimated (over 800 ppm) lithium concentrations in evaporation cuts. Direct Lithium Extraction (DLE) technologies, leveraging adsorbents like aluminum- and titanium-based materials, offer a transformative solution by achieving recovery rates of 70-90% in hours or days, significantly outperforming conventional methods while minimizing environmental impacts like water consumption and land use. The momentum behind DLE, with production reaching 83.7 kt LCE in 2023 and projected to hit 301 kt LCE by 2028, highlights

its potential to disrupt traditional supply chains and tap into unconventional sources like geothermal and oilfield brines.

However, the presence of magnesium and calcium in brines poses substantial challenges to DLE efficiency, reducing lithium recovery by 10-30% due to competition for adsorption sites, formation of complexes, fouling, scaling, and desorption complications, particularly in low-concentration brines (<200 mg/L Li) with high Mg/Li or Ca/Li ratios. The proposed methodology of pre-treating brine to remove divalent cations like Mg^{2+} and Ca^{2+} , as demonstrated in laboratory experiments with >99% purity salts (e.g., $CaCO_3$, $MgCO_3$), enhances lithium selectivity and reduces energy costs by up to 30% in subsequent DLE processes, though it introduces new risks of lithium loss to sodium crystals during evaporation. To optimize recovery from low-concentration brines, after separating divalent ions as commercial products, one needs to integrate membrane technologies to separate Li^+ from Na^+ , alongside innovations in nanostructured adsorbents and hybrid systems powered by renewable energy, which is crucial. Addressing these technical, economic, and environmental challenges through continued research, government support, and tailored process optimization will be essential to ensure DLE meets the sustainable lithium demands of the global energy transition. We are proposing an industry-first circular economy through a zero mineral discharge and zero liquid discharge process that also helps in the lithium recovery process for brine with a Li^+ concentration above 50 ppm concentration. One needs to select the lithium extraction process based on selectivity versus concentrating brine to increase the lithium concentration. In the presence of brine that is mainly Li and Na, with an extremely high Na/Li ratio, it is advisable to use the membrane method to recover lithium, provided economics are favorable. Further research is in progress to recover lithium from the sodium-rich brine without evaporating the brine.

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